

Unit Railroad

Deciding whether a conversion is set up correctly before any arithmetic happens, by reading the units instead of the number.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/unit-railroad/

1 Before you open anything

Answer from what you already know, with the page closed. The point is to have a commitment on paper that the demonstration can later confirm or overturn.

PREDICTION

A car is traveling 55.0 mi/hr. Using $1 \text{ mi} = 1.609344 \text{ km}$, write the same speed in m/s to three significant figures.

Now the part that matters: you had to deal with hours and seconds. Say in one sentence how you decided whether to multiply or divide by 3,600, and what told you that was the right choice.

2 One car, two orientations

Open the demonstration. In the **Problem** panel choose **1. km → m**. The shaded car on the left is the measurement you were given; the dashed car on the right is what the chain currently produces.

1. Before touching the bank, read the two boxes under the rail: **Units surviving** and **Number**. Record them in row 1.
2. In the **Conversion factor bank**, click **1 km / 1,000 m**. Do not flip it yet. Record row 2, including the tag printed on the dashed car.
3. Press the flip button (the up-down arrows) on that car. Record row 3.

Chain	Units surviving	Number	Tag on the dashed car
Nothing added yet			
1 km / 1,000 m, as the bank gives it			
the same car, flipped			

a. In row 2 the chain produced *some* number. Write one sentence explaining why that number is not an answer to the question that was asked.

b. In row 3, the km on the given car and the km on the factor are drawn struck through. What has to be true about where a unit sits for the demonstration to strike it out?

c. The **Number** box in row 3 does not read 2500. Read the caption under the rail and explain what it reads instead and why that form was chosen.

3 Which way up does a factor go?

Choose **3. mi/hr → m/s**. Build it one car at a time, and after every single action read the two boxes before doing anything else. Add the factors in the order listed; each one goes in unflipped first, then gets flipped.

1. With an empty chain, record row 1.
2. Add **1 mi / 1.609344 km**. Record row 2, then flip that car and record row 3. Leave it in whichever orientation cancels mi.
3. Add **1 km / 1,000 m**. Record row 4, then flip it and record row 5. Leave it in whichever orientation cancels km.
4. Add **1 hr / 3,600 s**. Record row 6, then flip it and record row 7.

Chain	Units surviving	Number	Tag on the dashed car
Nothing added yet			
+ 1 mi / 1.609344 km, as given			
that car flipped			
+ 1 km / 1,000 m, as given			
that car flipped			
+ 1 hr / 3,600 s, as given			
that car flipped			

a. Two of the three factors had to be flipped and one did not. Which one went in correctly as the bank supplied it, and what was different about where its unit started?

b. State the rule in your own words: given a unit you want to get rid of, how do you know which side of the new factor it belongs on? Do not use the words “flip” or “top and bottom” without saying *relative to what*.

c. Compare the finished answer with your prediction in Part 1. If the number matched, say whether your reasoning about the 3,600 matches the rule you just wrote. If it did not match, identify the single step where the two disagree.

Every factor in the bank is an **exact definition**, not a measurement, so none of them limits precision. The three significant figures in the answer come from the 55.0 mi/hr you were given, and from nothing else.

4 Two ways the setup can look right and not be

Both of these are shortcuts that work in other problems. Here is where each one stops working.

A. A cubed unit is not one unit. Choose **4. $\text{g}/\text{cm}^3 \rightarrow \text{kg}/\text{m}^3$** .

1. Add **1 m / 100 cm** and flip it so that cm sits on top. Record row 1.
2. Add the same factor a second time, flipped the same way. Record row 2.
3. Add it a third time, flipped the same way. Record row 3.
4. Now finish the conversion by adding **1 kg / 1,000 g** in whichever orientation cancels g. Record row 4.

Chain	Units surviving	Number	Tag on the dashed car
one 100 cm / 1 m			
two of them			
three of them			
plus the kg factor			

a. After row 1 the surviving units contained cm to a power. Which power, and why did one cm on top not remove cm^3 from the bottom?

b. Press **Clear chain** and rebuild the same conversion in two cars, using **$1 \text{ m}^3 / 1,000,000 \text{ cm}^3$** instead of three copies of the length factor. Does the **Number** box end up the same? Show where the 1,000,000 comes from, starting from $1 \text{ m} = 100 \text{ cm}$.

B. Cancelling units checks the setup, not the facts. Go back to **1. $\text{km} \rightarrow \text{m}$** and press **Clear chain**.

1. Open **Type your own factor**. Set the top to **1** and **km**, the bottom to **100** and **m**. Press **Add to bank**.

2. Click your new factor in the bank and flip it so the km cancels.

c. Record the **Units surviving**, the **Number**, and the tag on the dashed car.

d. The units cancelled perfectly and the answer is still wrong. State precisely what a complete cancellation does prove, and what it cannot prove.

The demonstration compares the number against an independently computed answer, which is why it can catch a false equality. On paper nothing does that for you — cancelled units mean your arrangement is right, and the equalities themselves are still yours to get right. Your typed factor stays in the bank for every problem until you reload the page.

5 Solve these without the demonstration

Close the page. Write every conversion factor as a fraction and show the cancellations before you touch a calculator. Then reopen the demonstration and check — if a number disagrees, find the car where the two setups differ.

1. Convert 2.50 km to inches, using $1 \text{ in} = 2.54 \text{ cm}$ exactly. Give the answer to three significant figures. (Check against problem 2.)

2. A car gets 32.0 mi/gal. Using $1 \text{ mi} = 1.609344 \text{ km}$ and $1 \text{ gal} = 3.785411784 \text{ L}$, convert this to km/L to three significant figures. Say which of the two units you dealt with first and why the order did not change the answer. (Check against problem 5.)

3. Without computing anything, decide what units survive this chain:

$$2.50 \text{ km} \times (1 \text{ km} / 1,000 \text{ m}) \times (1 \text{ m} / 100 \text{ cm}) \times (2.54 \text{ cm} / 1 \text{ in})$$

Then state the smallest change that would make it produce inches.

4. Explain, without doing any arithmetic, why both $1 \text{ mi} / 1.609344 \text{ km}$ and $1.609344 \text{ km} / 1 \text{ mi}$ may be used in a conversion chain, and what that fact means about the effect each one has on the *quantity* as opposed to the *units*.

6 On your own

Nothing here is graded, and the demonstration holds more than this worksheet used.

- Redo problem 3 without the mi–km factor, going $\text{mi} \rightarrow \text{ft} \rightarrow \text{in} \rightarrow \text{cm} \rightarrow \text{m}$ instead. Count the cars, then compare the final number with the short route. Should the two agree exactly, or only to three significant figures?
- Get a chain deliberately stuck, then press **What has to cancel next?** It names one unit and one side. Work out how it could know that without knowing the answer.
- Problem 4 offers $1 \text{ mL} / 1 \text{ cm}^3$ and $1 \text{ L} / 1,000 \text{ mL}$, which the shortest route never needs. Build a chain that passes through litres on the way and still lands on kg/m^3 . Then say what makes the $\text{mL}-\text{cm}^3$ factor unlike every other factor in the bank, and what that implies about the detour you just took.
- Type a factor with a unit that appears nowhere in the problem and add it to a finished chain. Predict what the two boxes will say before you look.
